

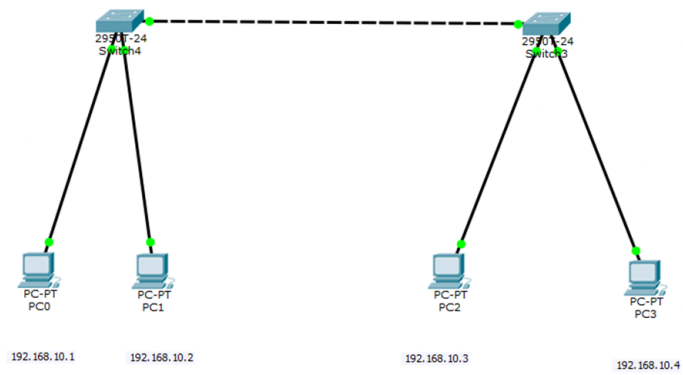


Programme	M.Tech.	Semester	Win'25-26
Course	Computer Networks	Code	MACSE513
Faculty	Dr. R. Kumar	Slot	L11+L12

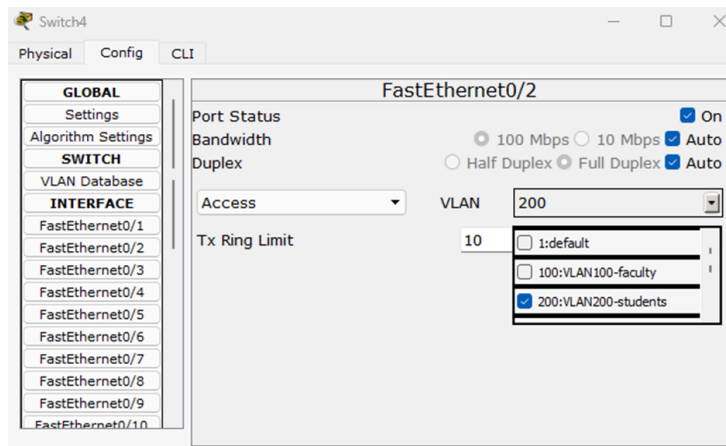
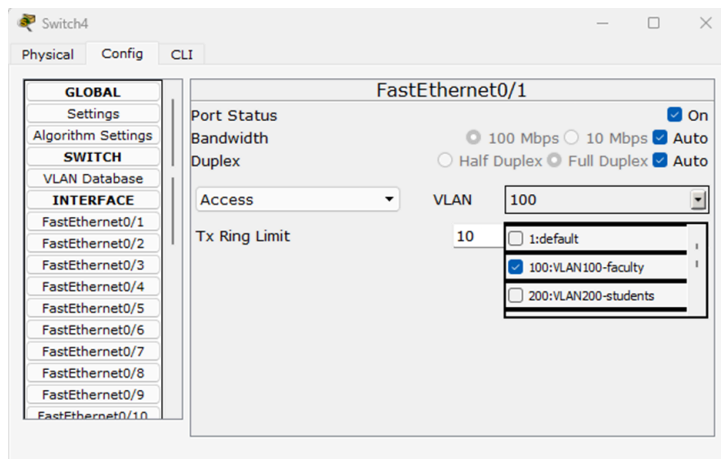
LAB: 7 - Virtual LAN

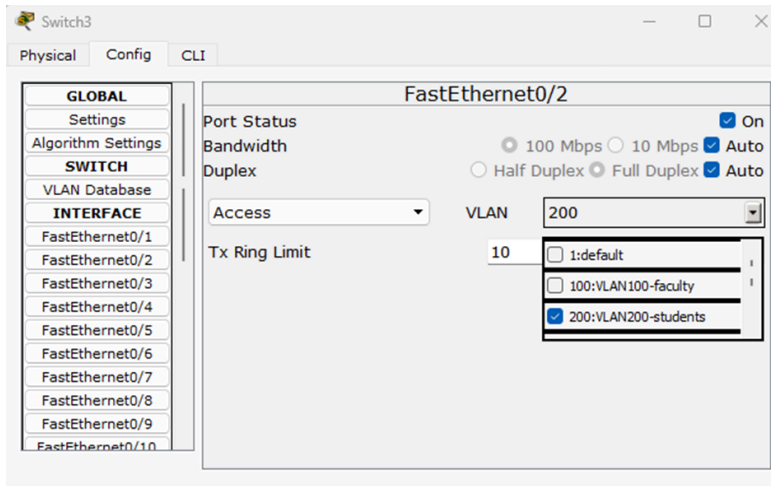
Submitted By: 25MCB1010, Patsamatla Bhaavya Ramita

1. Create a network of two manageable L2 switches as shown in Figure 1. Create one VLAN-100 for the faculty and another VLAN-200 for the students. Ensure that machines and devices in same VLAN can ping each other.
 - I created a network with two manageable Layer 2 switches.
 - On both switches, I made:
 - VLAN 100 is Faculty
 - VLAN 200 is Students
 - PCs for faculty were assigned to VLAN 100, and PCs for students were assigned to VLAN 200. Used the **switchport access vlan** command on each port to put the PCs in the right VLAN.
 - To connect the two switches, I set up a trunk link so VLAN traffic could pass between them.
 - Finally, I tested with ping: machines in VLAN 100 could ping each other, and machines in VLAN 200 could ping each other.



Configuring switches 4 and 3





```

PC>ping 192.168.10.1

Pinging 192.168.10.1 with 32 bytes of data:

Reply from 192.168.10.1: bytes=32 time=4ms TTL=128
Reply from 192.168.10.1: bytes=32 time=1ms TTL=128

Ping statistics for 192.168.10.1:
    Packets: Sent = 2, Received = 2, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 4ms, Average = 2ms

Control-C
^C
PC>ping 192.168.10.2

Pinging 192.168.10.2 with 32 bytes of data:

Ping statistics for 192.168.10.2:
    Packets: Sent = 1, Received = 0, Lost = 1 (100% loss),

Control-C
^C
PC>ping 192.168.10.3

Pinging 192.168.10.3 with 32 bytes of data:

Reply from 192.168.10.3: bytes=32 time=2ms TTL=128
Reply from 192.168.10.3: bytes=32 time=0ms TTL=128

Ping statistics for 192.168.10.3:
    Packets: Sent = 2, Received = 2, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 2ms, Average = 1ms

Control-C
^C
PC>ping 192.168.10.4

Pinging 192.168.10.4 with 32 bytes of data:

Ping statistics for 192.168.10.4:
    Packets: Sent = 1, Received = 0, Lost = 1 (100% loss),

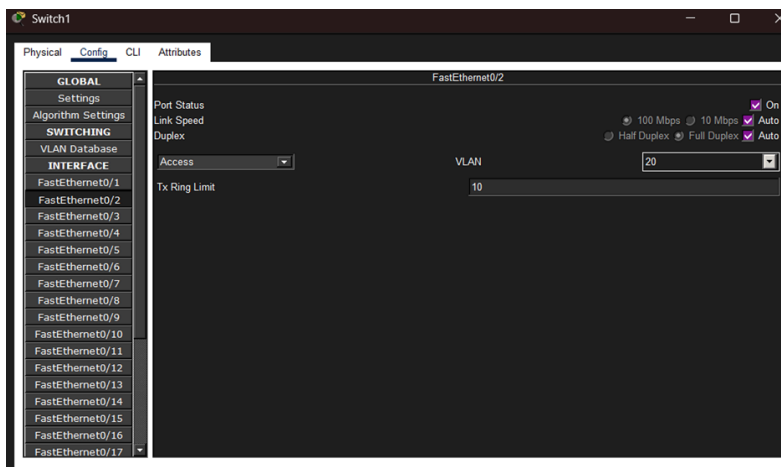
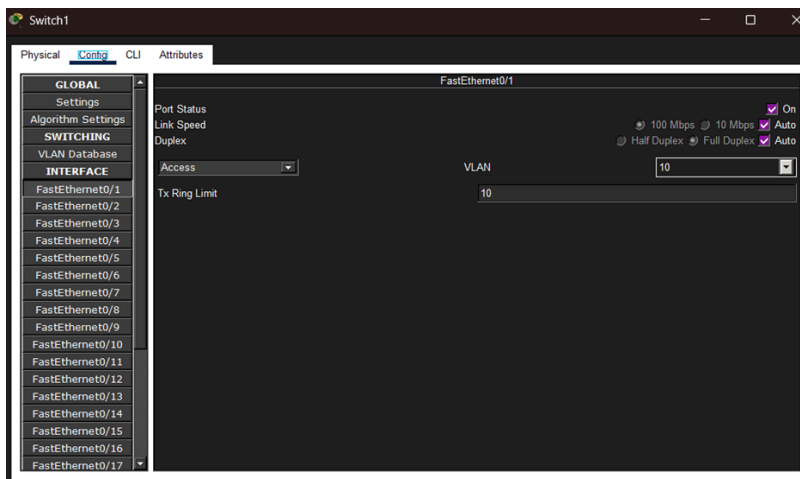
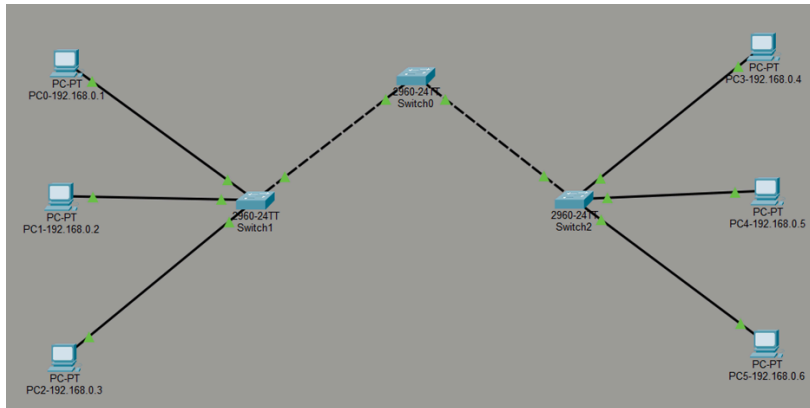
Control-C
^C
PC>|

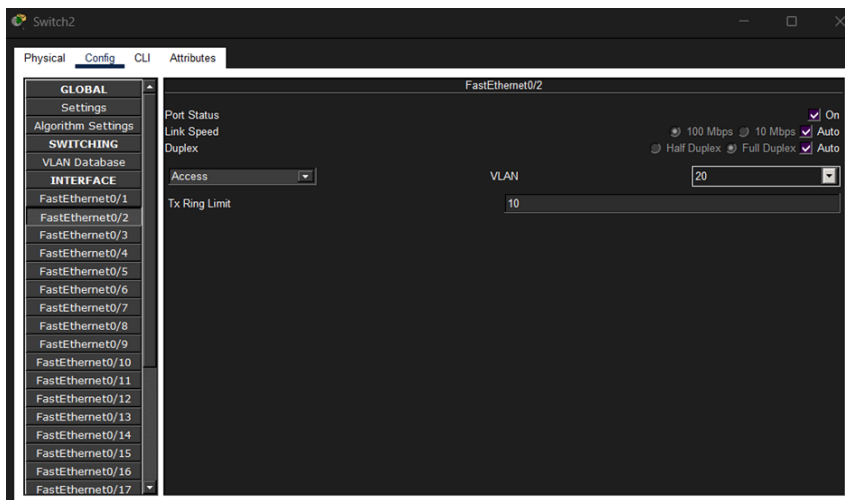
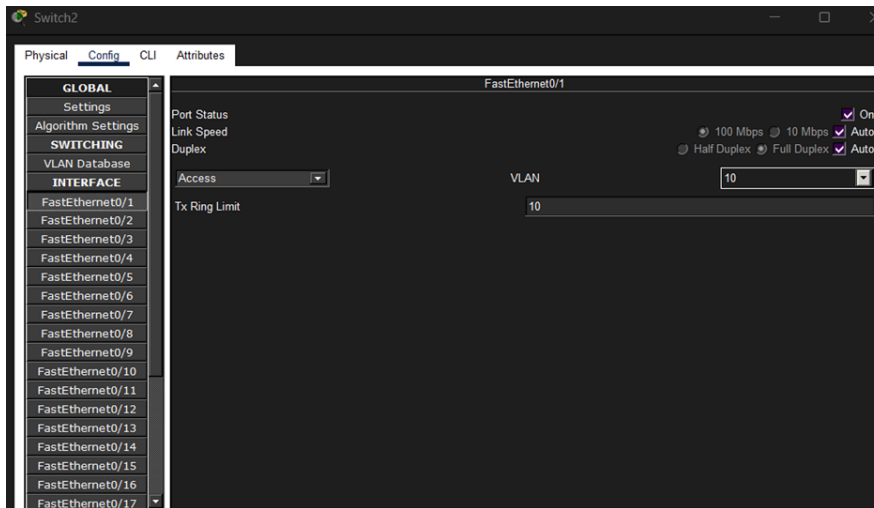
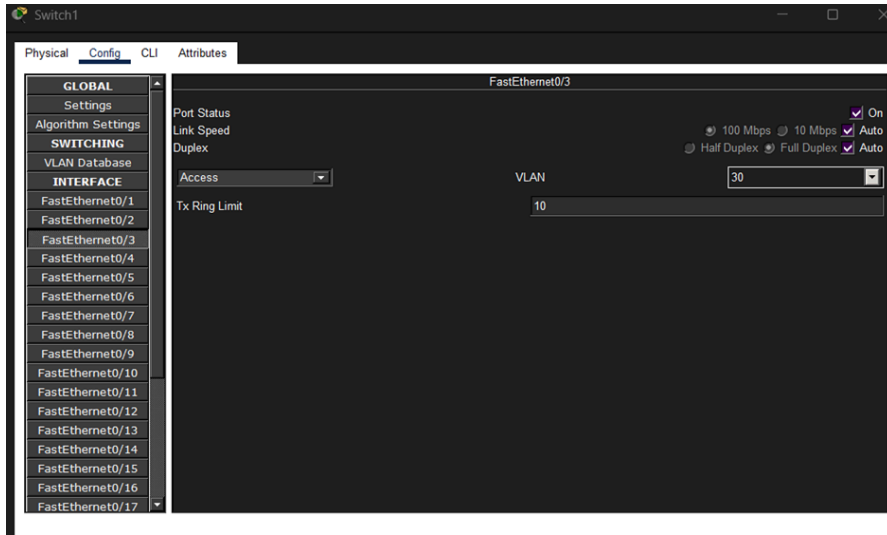
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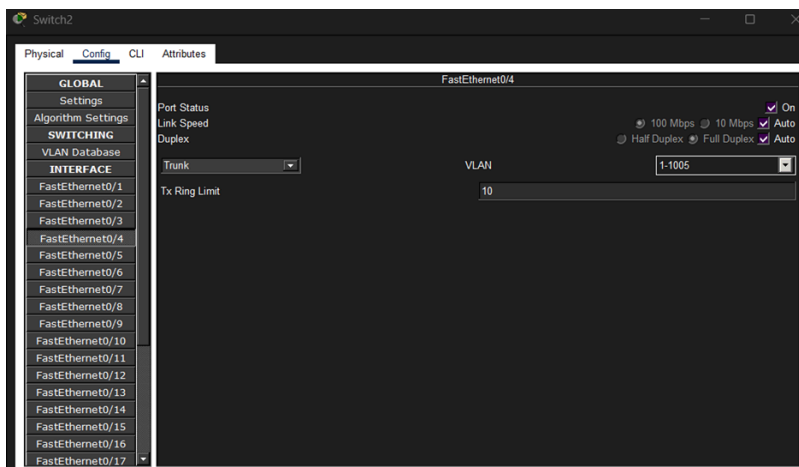
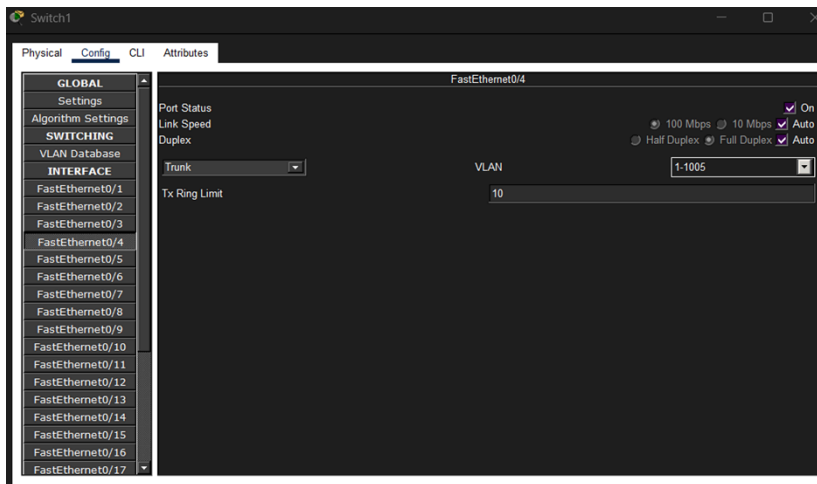
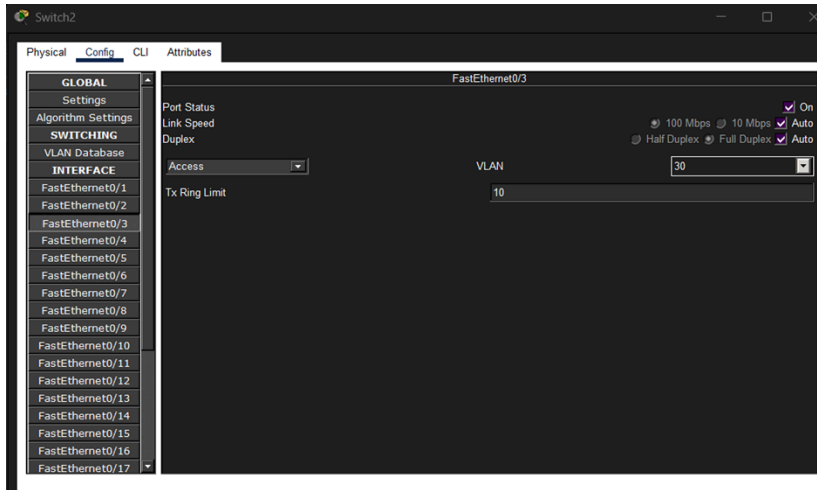
2. Create a topology in Figure.2 with three VLANs and ensure that machines and devices in same VLAN can ping each other.

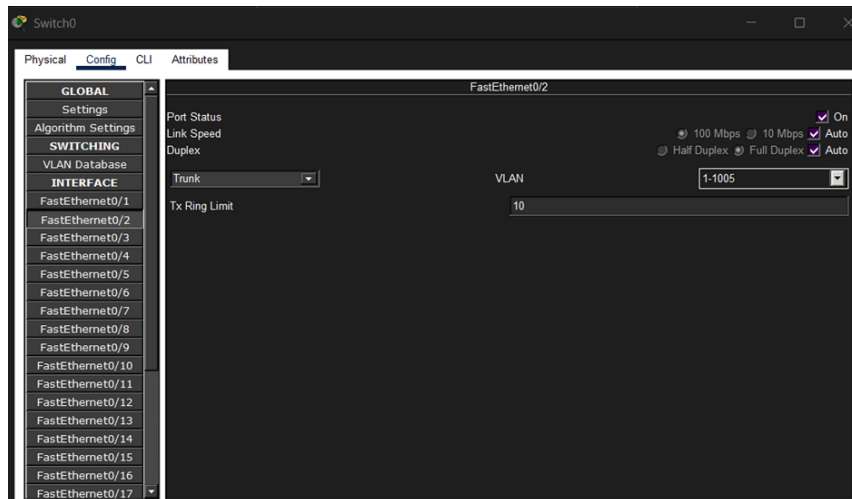
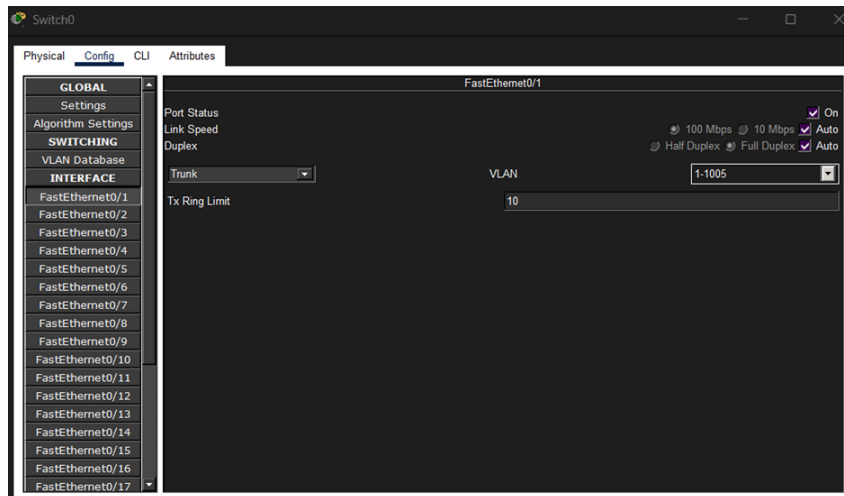
- In this topology, I used three Layer 2 switches.
- I created three VLANs:
 - VLAN 10 = Admin
 - VLAN 20 = Faculty
 - VLAN 30 → Students
- PCs were connected and assigned to their respective VLANs.

- Between the switches, I configured trunk ports so all VLANs could communicate across the switches.
- Again, tested with ping: devices in the same VLAN could reach each other, but devices in different VLANs stayed isolated.










```
PC0-192.168.0.1
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.0.3
Pinging 192.168.0.3 with 32 bytes of data:
Request timed out.
Ping statistics for 192.168.0.3:
    Packets: Sent = 2, Received = 0, Lost = 2 (100% loss),
Control-C
^C
C:\>ping 192.168.0.4
Pinging 192.168.0.4 with 32 bytes of data:
Reply from 192.168.0.4: bytes=32 time=1ms TTL=128
Reply from 192.168.0.4: bytes=32 time=1ms TTL=128
Reply from 192.168.0.4: bytes=32 time=1ms TTL=128
Reply from 192.168.0.4: bytes=32 time=1ms TTL=128
Ping statistics for 192.168.0.4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\>ping 192.168.0.5
Pinging 192.168.0.5 with 32 bytes of data:
Request timed out.
Ping statistics for 192.168.0.5:
    Packets: Sent = 1, Received = 0, Lost = 1 (100% loss),
Control-C
^C
C:\>ping 192.168.0.6
Pinging 192.168.0.6 with 32 bytes of data:
Request timed out.
Ping statistics for 192.168.0.6:
    Packets: Sent = 1, Received = 0, Lost = 1 (100% loss),
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PC1-192.168.0.2
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.0.4
Pinging 192.168.0.4 with 32 bytes of data:
Request timed out.
Ping statistics for 192.168.0.4:
    Packets: Sent = 1, Received = 0, Lost = 1 (100% loss),
Control-C
^C
C:\>ping 192.168.0.5
Pinging 192.168.0.5 with 32 bytes of data:
Reply from 192.168.0.5: bytes=32 time=1ms TTL=128
Reply from 192.168.0.5: bytes=32 time=1ms TTL=128
Reply from 192.168.0.5: bytes=32 time=1ms TTL=128
Reply from 192.168.0.5: bytes=32 time=1ms TTL=128
Ping statistics for 192.168.0.5:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\>ping 192.168.0.6
Pinging 192.168.0.6 with 32 bytes of data:
Request timed out.
Ping statistics for 192.168.0.6:
    Packets: Sent = 1, Received = 0, Lost = 1 (100% loss),
Control-C
^C
C:\>ping 192.168.0.1
Pinging 192.168.0.1 with 32 bytes of data:
Request timed out.
Ping statistics for 192.168.0.1:
    Packets: Sent = 1, Received = 0, Lost = 1 (100% loss),
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PC2-192.168.0.3
Physical Config Desktop Programming Attributes
Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.0.4

Pinging 192.168.0.4 with 32 bytes of data:

Request timed out.

Ping statistics for 192.168.0.4:
    Packets: Sent = 1, Received = 0, Lost = 1 (100% loss),

Control-C
^C
C:\>ping 192.168.0.5

Pinging 192.168.0.5 with 32 bytes of data:

Request timed out.

Ping statistics for 192.168.0.5:
    Packets: Sent = 1, Received = 0, Lost = 1 (100% loss),

Control-C
^C
C:\>ping 192.168.0.6

Pinging 192.168.0.6 with 32 bytes of data:

Reply from 192.168.0.6: bytes=32 time<1ms TTL=128
Reply from 192.168.0.6: bytes=32 time<1ms TTL=128
Reply from 192.168.0.6: bytes=32 time<1ms TTL=128
Reply from 192.168.0.6: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.0.6:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.0.2

Pinging 192.168.0.2 with 32 bytes of data:

Request timed out.

Ping statistics for 192.168.0.2:
    Packets: Sent = 1, Received = 0, Lost = 1 (100% loss),
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